

CS202 Seminar - Decorator Design Pattern

Agenda

- 1 Motivation
- 2 Naive Solution
- 3 Problems
- 4 Decorator Pattern
- 5 Class Diagram
- 6 Implementation
- 7 Advantages and Disadvantages
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- 9 Facade Pattern
- 10 Relationships with Other Design Patterns
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1. A Real-world Problem

Imagine your team is building an order management software for a coffee shop chain (e.g., The Coffee House).

Available coffee:

- Dark Roast
- House Blend
- Espresso

Customers rarely order plain coffee.

They often request additional condiments:

- Milk
- Sugar
- Caramel

Each topping changes

- Price
- Description

2. Naive Solution - Inheritance

The first idea is using inheritance.

Example:

- Beverage
- DarkRoast
- DarkRoastWithMilk
- DarkRoastWithMilkAndSugar
- DarkRoastWithMilkAndSugarAndWhip
- EspressoWithWhip
- HouseBlendWithMilk
- ...

2. Naive Solution - Inheritance

The first idea is using inheritance.

Example:

- Beverage
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- DarkRoastWithMilk
- DarkRoastWithMilkAndSugar
- DarkRoastWithMilkAndSugarAndWhip
- EspressoWithWhip
- HouseBlendWithMilk
- ...

Problem

Every combination requires a new class.

Naive Solution - Boolean Flags

Another approach is storing every topping inside the Beverage class.

```
1 class Beverage {  
2  
3 private:  
4  
5     bool hasMilk;  
6     bool hasSugar;  
7     bool hasWhip;  
8     bool hasCaramel;  
9  
10 public:  
11     ...  
12 };
```

Naive Solution - Boolean Flags

Another approach is storing every topping inside the Beverage class.

```
1 class Beverage {  
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3 private:  
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5     bool hasMilk;  
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7     bool hasWhip;  
8     bool hasCaramel;  
9  
10 public:  
11     ...  
12 };
```

Everything is controlled by boolean variables.

3. Problems with the Naive Solution

Class Explosion

4 coffee types $\times$ many toppings = dozens (or hundreds) of subclasses.

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Hard Maintenance

Changing the price of milk requires modifying many classes.

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Changing the price of milk requires modifying many classes.

Violates Open/Closed Principle

Adding a new topping (e.g. Boba) requires modifying Beverage.

3. Problems with the Naive Solution

Class Explosion

4 coffee types $\times$ many toppings = dozens (or hundreds) of subclasses.

Hard Maintenance

Changing the price of milk requires modifying many classes.

Violates Open/Closed Principle

Adding a new topping (e.g. Boba) requires modifying Beverage.

No Runtime Flexibility

Cannot naturally support

- Dynamic combinations

4. Decorator Pattern

Decorator is a **Structural Design Pattern**.

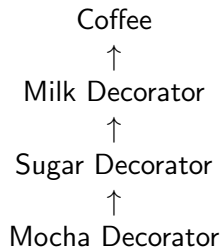
Instead of inheritance,
we dynamically attach new behaviors to an object.

Core philosophy:

Composition over Inheritance

Core Concepts

Decorator uses wrapper objects.



5. General Structure of Decorator Pattern

Main Components

- **Component**: Common interface for all objects.
- **ConcreteComponent**: The original object.
- **Decorator**: Abstract wrapper containing a Component.
- **ConcreteDecorator**: Adds new behavior while preserving the interface.

Flow:

Component $\leftarrow$ *Decorator* $\leftarrow$ *Decorator* $\leftarrow$ *Decorator*

Every decorator is still recognized as a Component.

Coffee UML (Mermaid)

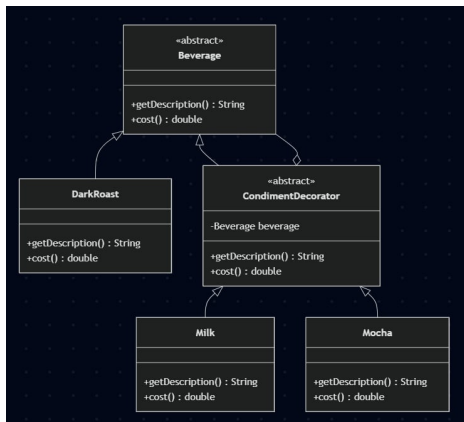


Figure: Coffee UML diagram

7. Component

```
1 class Beverage {  
2 public:  
3     virtual string getDescription() const = 0;  
4     virtual double cost() const = 0;  
5     virtual ~Beverage() = default;  
6 };
```


7. Component

```
1 class Beverage {  
2 public:  
3     virtual string getDescription() const = 0;  
4     virtual double cost() const = 0;  
5     virtual ~Beverage() = default;  
6 };
```

Role

Defines the common interface. Every coffee and every topping must inherit from Beverage.

Concrete Component

```
1 class DarkRoast : public Beverage {  
2 public:  
3     string getDescription() const override {  
4         return "Dark_Roast_Coffee";  
5     }  
6  
7     double cost() const override {  
8         return 30.0;  
9     }  
10 };
```

Concrete Component

```
1 class DarkRoast : public Beverage {  
2 public:  
3     string getDescription() const override {  
4         return "Dark_Roast_Coffee";  
5     }  
6  
7     double cost() const override {  
8         return 30.0;  
9     }  
10 };
```

DarkRoast is the original object before decoration.

Base Decorator

```
1 class CondimentDecorator : public Beverage {  
2 protected:  
3     Beverage* beverage;  
4 public:  
5     CondimentDecorator(Beverage* bev) : beverage(bev) {}  
6  
7     virtual ~CondimentDecorator() {  
8         delete beverage;  
9     }  
10 };
```

Base Decorator

```
1 class CondimentDecorator : public Beverage {  
2 protected:  
3     Beverage* beverage;  
4 public:  
5     CondimentDecorator(Beverage* bev) : beverage(bev) {}  
6  
7     virtual ~CondimentDecorator() {  
8         delete beverage;  
9     }  
10 };
```

The decorator wraps another Beverage object.
Every decorator stores a pointer to another Beverage.

Concrete Decorator: Milk

```
1 class Milk : public CondimentDecorator {  
2 public:  
3     Milk(Beverage* bev) : CondimentDecorator(bev) {}  
4  
5     string getDescription() const override {  
6         return beverage->getDescription() + ", Milk";  
7     }  
8  
9     double cost() const override {  
10        return beverage->cost() + 5.0;  
11    }  
12};
```

Concrete Decorator: Milk

```
1 class Milk : public CondimentDecorator {  
2 public:  
3     Milk(Beverage* bev) : CondimentDecorator(bev) {}  
4  
5     string getDescription() const override {  
6         return beverage->getDescription() + ", Milk";  
7     }  
8  
9     double cost() const override {  
10        return beverage->cost() + 5.0;  
11    }  
12};
```

Concrete Decorator: Mocha

```
1 class Mocha : public CondimentDecorator {  
2 public:  
3     Mocha(Beverage* bev) : CondimentDecorator(bev) {}  
4     string getDescription() const override { return beverage->  
5         getDescription() + ", Mocha";  
6     }  
7     double cost() const override {  
8         return beverage->cost() + 8.0;  
9     }  
10 };
```


Concrete Decorator: Mocha

```
1 class Mocha : public CondimentDecorator {
2 public:
3     Mocha(Beverage* bev) : CondimentDecorator(bev) {}
4     string getDescription() const override { return beverage->
        getDescription() + ", Mocha";
5     }
6
7     double cost() const override {
8         return beverage->cost() + 8.0;
9     }
10 };
```

Every new decorator extends the previous Beverage.
No existing class needs modification.

Client Code

```
1 int main() {  
2     Beverage* myDrink = new DarkRoast();  
3  
4     myDrink = new Milk(myDrink);  
5     myDrink = new Mocha(myDrink);  
6  
7     cout << myDrink->getDescription() << endl;  
8     cout << myDrink->cost();  
9     delete myDrink;  
10 }
```

How Decoration Works

The object is wrapped layer by layer.

DarkRoast



Milk(DarkRoast)



Mocha(Milk(DarkRoast))

How Decoration Works

The object is wrapped layer by layer.

DarkRoast



Milk(DarkRoast)



Mocha(Milk(DarkRoast))

Every wrapper is still a Beverage.

Execution Flow

Calling

myDrink - $\rightarrow$ *cost()*

starts from the outermost decorator.

Execution order:

Mocha $\rightarrow$ *Milk* $\rightarrow$ *DarkRoast*

Then the values are returned back.

$$8 + 5 + 30 = 43$$

Program Output

Description

Dark Roast Coffee, Milk, Mocha

Total Cost

43k VND

Advantages

Runtime Flexibility

Features can be added dynamically while the program is running.

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Reusable Components

Each decorator can be reused independently.

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Runtime Flexibility

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Supports any topping combination without creating new subclasses.

Reusable Components

Each decorator can be reused independently.

Follows SOLID

- Single Responsibility Principle
- Open/Closed Principle

Disadvantages

Many Small Classes

Each feature requires its own decorator class.

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Harder Debugging

Many wrapper layers produce long call stacks.

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Complex Initialization

Nested construction may become difficult.

Example:

```
new Mocha(new Milk(new Mocha(new DarkRoast()))))
```

Strategy Pattern

Core Idea

Encapsulate different algorithms behind a common interface and allow them to be switched at runtime.

- One task → Multiple possible algorithms
- Each algorithm is implemented in a separate class
- The algorithm can be changed dynamically at runtime
- Avoids large if-else or switch-case blocks

Example: A graph visualizer supporting multiple pathfinding algorithms.

The Problem: Without Strategy

```
1 class GraphVisualizer {  
2 public:  
3     void RunPathfinding(string algoType) {  
4         if (algoType == "Dijkstra") {  
5             }  
6         else if (algoType == "BellmanFord") {  
7             }  
8         else if (algoType == "BFS") {  
9             }  
10    }  
11 };
```

Problems

- The class becomes very large and difficult to maintain
- Adding a new algorithm requires modifying existing code
- High risk of introducing bugs

The Solution: Strategy Pattern

```
1 class IPathfindingStrategy {  
2 public:  
3     virtual void ExecuteAlgorithm() = 0;  
4     virtual ~IPathfindingStrategy() {}  
5 };
```

Common Interface

Every pathfinding algorithm follows the same contract.

- DijkstraStrategy
- BellmanFordStrategy
- BFSStrategy
- ...

Each algorithm is encapsulated in its own class. The main application does not need to know the implementation details.

Concrete Strategies

```
1 class DijkstraStrategy : public IPathfindingStrategy {
2 public:
3     void ExecuteAlgorithm() override { // Dijkstra's algorithm
4     }
5 };
6
7 class BellmanFordStrategy : public IPathfindingStrategy {
8 public:
9     void ExecuteAlgorithm() override { // Bellman-Ford algorithm
10    }
11 };
```

Key Idea

Different algorithms are interchangeable because they implement the same interface.

Context: Graph Visualizer

```
1 class GraphVisualizer {  
2 private:  
3     IPathfindingStrategy* strategy;  
4 public:  
5     void SetStrategy(IPathfindingStrategy* s) {  
6         strategy = s;  
7     }  
8     void Run() {  
9         if (strategy)  
10            strategy->ExecuteAlgorithm();  
11    }  
12};
```

Context

The GraphVisualizer does not care which algorithm is used. It simply executes the selected strategy.

Switching Strategies at Runtime

```
1 GraphVisualizer app;  
2  
3 // User selects Dijkstra  
4 app.SetStrategy(new DijkstraStrategy());  
5 app.Run();  
6  
7 // User switches to Bellman-Ford  
8 app.SetStrategy(new BellmanFordStrategy());  
9 app.Run();
```

Result

The algorithm changes, but the GraphVisualizer code remains unchanged.

One task, multiple interchangeable algorithms

- Encapsulate each algorithm separately
- Use a common interface
- Switch algorithms at runtime
- Reduce complex conditional logic
- Easy to extend with new strategies

Main Benefit

Add a new algorithm without modifying the existing context.

Facade Pattern

Core Idea

Provide a simple, unified interface to a complex subsystem.

Complex System $\longrightarrow$ **Simple Interface**

- Hide unnecessary implementation details
- Reduce the complexity seen by the client
- Provide a single entry point to a subsystem

Example: Starting a computer by pressing the power button.

The Problem: A Complex Subsystem

```
1 class CPU {
2 public:
3     void Freeze() {}
4     void Jump(long position) {}
5     void Execute() {}
6 };
7 class Memory {
8 public:
9     void Load(long position, char* data) {}
10 };
11 class HardDrive {
12 public:
13     char* Read(long lba, int size) {
14         return "OS_data";
15     }
16 };
```

The Problem: A Complex Subsystem (cont.)

Problem

Starting a computer requires coordinating multiple components: CPU, RAM, and Hard Drive.

The Solution: Computer Facade

```
1 class ComputerFacade {  
2 private:  
3     CPU cpu;  
4     Memory ram;  
5     HardDrive hdd;  
6  
7 public:  
8     void StartComputer() {  
9         cpu.Freeze();  
10        ram.Load(0x00, hdd.Read(0, 1024));  
11        cpu.Jump(0x00);  
12        cpu.Execute();  
13    }  
14 };
```


Facade Pattern

Facade

ComputerFacade hides the complex startup process behind a single method: `StartComputer()`.

Client: Simple Interface

```
1 int main() {  
2     ComputerFacade myPC;  
3  
4     // Press the power button  
5     myPC.StartComputer();  
6  
7     return 0;  
8 }
```

One method call. That's it.

- The client does not interact directly with CPU, RAM, or HDD
- Internal complexity is hidden
- The client only depends on the Facade

Press the Power Button

`StartComputer()`

- CPU initialization
- Loading data from the Hard Drive
- Loading data into Memory
- Starting CPU execution

The User

Only needs to press one button. They do not need to understand how the internal components work together.

Decorator, Facade, and Strategy solve different problems

Decorator

Add Behavior

Extends an object's functionality without modifying its original class.

Facade

Hide Complexity

Provides a simple interface to a complex subsystem.

Strategy

Change Behavior

Allows different algorithms to be switched at runtime.

Decorator → *What can I add?*

Facade → *How can I simplify access?*

Strategy → *How can I change the algorithm?*

Decorator, Facade, and Strategy

Pattern	Main Purpose	Key Question
Decorator	Add or extend behavior	<i>How can I add functionality?</i>
Facade	Simplify access to a complex system	<i>How can I make the system easier to use?</i>
Strategy	Switch between different algorithms	<i>How can I change the way a task is performed?</i>

The Big Picture

All three patterns help reduce coupling and improve flexibility, but they focus on different aspects of software design.

Example: A Pathfinding System

Solution

The client calls one simple method through the Facade. The Strategy selects the pathfinding algorithm, while Decorators add extra functionality.

Decorator

Adds new behavior to an object.

Facade

Hides system complexity behind a simple interface.

Strategy

Allows different algorithms to be exchanged easily.

Key Takeaway

Decorator adds behavior, Facade simplifies access, Strategy changes behavior.

9. Real-World Applications: Java I/O Streams

Conceptual Background

In the `java.io` package, reading data can originate from multiple sources (files, network sockets, byte arrays) and require various processing capabilities (buffering, decompression, data parsing). Creating subclass combinations like `BufferedGzipFileInputStream` would cause severe class explosion.

9. Real-World Applications: Java I/O Streams

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GoF Design Pattern Mapping

- **Component:** `InputStream` (Abstract base interface defining `read()`)
- **Concrete Component:** `FileInputStream` (Reads raw bytes directly from a file)
- **Base Decorator:** `FilterInputStream` (Holds an internal `InputStream` reference)
- **Concrete Decorators:** `BufferedInputStream`, `GZIPInputStream` (Add buffering and decompression)

Java I/O Streams: Execution Flow

```
1 // Base file reader wrapped with buffering and GZIP decompression
2 InputStream fileStream = new FileInputStream("data.txt.gz");
3 InputStream bufferedStream = new BufferedInputStream(fileStream);
4 InputStream gzipStream = new GZIPInputStream(bufferedStream);
5
6 int data = gzipStream.read(); // Execution propagates down through
   wrappers
```

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How Decoration Works at Runtime

- 1 The client calls `read()` on the outermost decorator (`GZIPInputStream`).
- 2 `GZIPInputStream` requests raw compressed chunks from `BufferedInputStream`.
- 3 `BufferedInputStream` fetches byte blocks from `FileInputStream`.
- 4 Data flows back up through the stack, being buffered and decompressed dynamically!

Architectural Problem

Modern backend frameworks (Express.js, ASP.NET Core, Spring Filters) must process incoming HTTP requests with cross-cutting concerns like authentication, CORS headers, logging, and error handling before reaching the final controller.

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The Middleware Wrapper Concept

Instead of polluting every controller endpoint with repetitive security checks, middleware functions act as an onion-skin wrapper stack around the business logic controller.



Backend Middleware: Code & Delegation

```
1 // Conceptual Express.js / ASP.NET Middleware Stack
2 void AuthMiddleware(Request req, Response res, NextFunction next) {
3     if (!req.isValidToken()) {
4         res.sendError(401, "Unauthorized");
5         return; // Intercept execution early
6     }
7     next(); // Pass control to the next decorated layer in the
           pipeline
8 }
```

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Why Decorator Pattern Fits Here

- **Dynamic Stacking:** Middleware layers can be added or removed per route without modifying the target route controller.
- **Pre/Post Processing:** A middleware layer can execute code both *before* calling `next()` and *after* `next()` returns (e.g., measuring response latency).

Python Decorators

Native Language Support

Python integrates the Decorator pattern directly into its language syntax via the `@decorator_name` annotation. In Python, functions are first-class objects, allowing higher-order functions to wrap target callables dynamically.

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Syntactic Sugar Mechanism

Writing an annotation in Python:

```
@login_required  
def view_dashboard(): ...
```

is exact syntactic sugar for wrapping the function object at definition time:

```
view_dashboard = login_required(view_dashboard)
```

Python Decorators: Practical Implementation

```
1 def login_required(func):
2     def wrapper(*args, **kwargs):
3         if not current_user.is_authenticated:
4             raise PermissionError("User_must_log_in_to_view_page")
5         return func(*args, **kwargs) # Delegate to original function
6     return wrapper
7
8 @login_required
9 def view_dashboard():
10     return "Secret_User_Dashboard_Data"
```

Python Decorators: Practical Implementation

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6     return wrapper
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8 @login_required
9 def view_dashboard():
10     return "Secret User Dashboard Data"
```

Key Takeaway

The `login_required` wrapper adds authentication checks dynamically without modifying a single line of code inside `view_dashboard()`.

React Higher-Order Components (HOC)

UI Component Decoration

In modern web frontend development with React, user interface components should focus on rendering layout. Cross-cutting UI concerns (such as authorization checks, global theme injection, or data fetching) are attached using Higher-Order Components.

React Higher-Order Components (HOC)

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GoF Structural Mapping in React

- **Component:** Base React Component (`UserProfilePage`)
- **Decorator HOC:** Wrapper function (`withAuth`) returning an enhanced Component
- **Concrete Result:** `withAuth(UserProfilePage)`

React HOC: Code & Composition

```
1 // React Higher-Order Component Wrapper Example
2 const withAuth = (WrappedComponent) => {
3   return (props) => {
4     if (!props.isAuthenticated) {
5       return <LoginPage />;
6     }
7     return <WrappedComponent {...props} />;
8   };
9 };
10
11 // Usage: Create an authenticated view without touching
12 // UserProfilePage source code
13 const ProtectedProfile = withAuth(UserProfilePage);
```

React HOC: Code & Composition

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9 };
10
11 // Usage: Create an authenticated view without touching
12 // UserProfilePage source code
13 const ProtectedProfile = withAuth(UserProfilePage);
```

Benefit

Multiple HOC decorators can be composed: `withTheme(withAuth(UserProfilePage))`, giving UI components modular, reusable capabilities.

Flutter Widget Trees

Composition Over Inheritance in Mobile UI

Google's Flutter framework completely abandons traditional subclassing for UI elements. Instead of creating a `PaddedBorderedText` subclass, Flutter adopts the philosophy:

"Everything is a Widget".

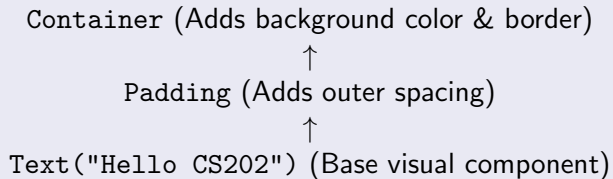
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Visual Decorator Tree

Base content is decorated visually by nesting wrapper widgets layer by layer:



Flutter Widgets: Code & Render Pipeline

```
1 // Flutter UI composition via nested decorator widgets
2 Widget build(BuildContext context) {
3     return Container(
4         color: Colors.blue,
5         child: Padding(
6             padding: EdgeInsets.all(16.0),
7             child: Text("Hello_␣CS202_␣Seminar"),
8         ),
9     );
10 }
```

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9     );
10 }
```

Render Execution Flow

During rendering, Flutter passes layout constraints down through the Container and Padding wrappers to the inner Text child. The computed pixel dimensions return back up the tree—exactly mirroring how decorator cost calculations bubble up in object wrapping!

Question 1

Decorator belongs to which Design Pattern category?

- A. Creational
- B. Structural
- C. Behavioral
- D. Architectural

Question 1

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Answer

B. Structural

Question 2

Which design principle does Decorator emphasize?

- A. Inheritance over Composition
- B. Composition over Inheritance
- C. Tight Coupling
- D. Procedural Programming

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Answer

B. Composition over Inheritance

Question 3

What is the main purpose of Decorator?

- A. Clone an object
- B. Restrict object creation
- C. Add new behavior at Runtime
- D. Simplify a subsystem

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Answer

C. Add new behavior at Runtime

Question 4

Which is a disadvantage of Decorator?

- A. Violates OCP
- B. Causes Class Explosion
- C. Long call stack and many small classes
- D. Cannot work in OOP languages

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Answer

C. Long call stack and many small classes

Question 5

How do we implement Double Milk?

- A. Create DoubleMilkDecorator
- B. Add count = 2
- C. Wrap Milk twice
- D. Copy the code twice

Question 5

How do we implement Double Milk?

- A. Create DoubleMilkDecorator
- B. Add count = 2
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Answer

C. Wrap Milk twice

Thank you for listening!